

Assignment 1: AI System Specification

Fraud Detection System for Digital Payments and Banking Transactions Course: Introduction to Artificial Intelligence | Topic: PEAS Framework and Task Environment Analysis

Task 1: PEAS Framework

Performance Measures

1. Fraud detection accuracy
2. False positive rate
3. Transaction processing speed
4. Financial loss prevention
5. Customer satisfaction

Environment

- Online banking platforms
- Mobile payment applications
- Credit/debit card transaction networks
- E-commerce websites
- Financial institutions and payment gateways

Actuators

Block suspicious transactions, Flag transactions for review, Send fraud alerts, Freeze accounts, Generate reports

- Block suspicious transactions
- Flag transactions for manual review
- Send fraud alerts to customers
- Freeze compromised accounts
- Generate fraud investigation reports

Sensors

Transaction records, Account activity logs, Device information, IP address, Geolocation data, Login behavior patterns

- Transaction records
- Customer account activity logs
- Device information
- IP address and geolocation data
- Login behavior patterns

Task 2: Environment Classification

Environment Dimension	Choice	Justification
Fully Observable vs. Partially Observable	Partially Observable	The system cannot access all information about a user's intentions or external fraud activities.
Single-agent vs. Multi-agent	Multi-agent	Fraudsters actively adapt their strategies, creating interactions between the detection system and malicious actors.
Deterministic vs. Stochastic	Stochastic	Transaction outcomes and fraud patterns involve uncertainty and randomness.
Episodic vs. Sequential	Sequential	Previous transactions influence future fraud detection decisions.
Static vs. Dynamic	Dynamic	Fraud techniques and transaction behaviors continuously change in real time.
Discrete vs. Continuous	Continuous	Transaction data such as amounts, times, and locations change continuously.

Task 3: Critical Analysis

1. Greatest Technical Challenge

The **Dynamic** nature of the environment is the greatest challenge. Fraudsters constantly develop new methods to bypass security systems. Therefore, the AI model must be regularly updated and trained on new data to maintain high detection accuracy.

2. Utility Function

Utility Function:

$$U = 0.7 \times \text{DetectionAccuracy} - 0.3 \times \text{FalsePositiveRate}$$

If the weight of **Detection Accuracy** is doubled:

$$U = 1.4 \times \text{DetectionAccuracy} - 0.3 \times \text{FalsePositiveRate}$$

The system will prioritize catching more fraudulent transactions, even if it occasionally flags more legitimate transactions as suspicious.